

42–(MathAA/10_HL_Winter_2022_Q8)–Trigonometry, Algebra-functions-Roots

[Maximum mark: 5]

Let $f(x) = \cos(x - k)$, where $0 \leq x \leq a$ and $a, k \in \mathbb{R}$.

- (a) Consider the case where $k = \frac{\pi}{2}$.

By sketching a suitable graph, or otherwise, find the largest value of a for which the inverse function f^{-1} exists.

[2]

- (b) Find the largest value of a for which the inverse function f^{-1} exists in the case where $k = \pi$.

[1]

- (c) Find the largest value of a for which the inverse function f^{-1} exists in the case where $\pi < k < 2\pi$. Give your answer in terms of k .

[2]



1-(MAT/12_HL_Summer_2012_Q5)-Matrices

Three non-singular 2×2 matrices A , B and X satisfy $4A - 5BX = B$.

(a) Find X in terms of A and B . [2 marks]

(b) Given that $A = 2B$, find X . [3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



2-(MAT/10_HL_Winter_2012_Q6)-Matrices

Consider the following equations, where $a, b \in \mathbb{R}$:

$$x + 3y + (a - 1)z = 1$$

$$2x + 2y + (a - 2)z = 1$$

$$3x + y + (a - 3)z = b.$$

- (a) If each of these equations defines a plane, show that, for any value of a , the planes do not intersect at a unique point. [3 marks]

.....

.....

.....

.....

.....

.....

.....

.....



- (b) Find the value of b for which the intersection of the planes is a straight line. [4 marks]

.....

.....

.....

.....

.....

.....

.....

.....



3-(MAT/10_HL_Winter_2012_Q11)-Matrices

Consider the matrix $A = \begin{pmatrix} a & a-1 \\ b & b \end{pmatrix}$, $b \neq 0$.

(a) For $a = 2$ and $b = 1$, show that $(A^2 - 3A)^2 = I$. [3 marks]

(b) Find the value of a and the value of b in each of the following cases:

(i) $A \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$;

(ii) $A^{-1} \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ -2 \end{pmatrix}$. [6 marks]

(c) Consider the lines l_1 and l_2 defined by the equations $ax + (a-1)y = 0$ and $bx + by = 1$, respectively, where $b \neq 0$.

(i) Find the coordinates of the intersection point of the lines l_1 and l_2 in terms of a and b .

(ii) Given that the lines are perpendicular, find the coordinates of the point of intersection in terms of b . [9 marks]



4-(MAT/12_HL_Summer_2013_Q2)-Matrices

Consider the matrices $A = \begin{pmatrix} 5 & 3 \\ 3 & 2 \end{pmatrix}$, $B = \begin{pmatrix} 3 & -5 \\ 2 & -3 \end{pmatrix}$.

(a) Find $\det A$ and hence write down the matrix A^{-1} . [2 marks]

(b) Find the matrix $A^{-1}B$. [3 marks]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



5-(MAT/11_HL_Summer_2013_Q3)-Matrices

[Maximum mark: 5]

Given $A = \begin{pmatrix} 1 & 2 \\ 1 & 1 \end{pmatrix}$ and $B = \begin{pmatrix} 4 & -3 \\ 1 & 1 \end{pmatrix}$, find the matrix X , such that $AXA^{-1} = B$.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



6-(MAT/11_HL_Summer_2013_Q6)-Matrices

[Maximum mark: 6]

The matrix A is such that $A^2 = I$, where I is the identity matrix. Use mathematical induction to prove that $(A + I)^n = 2^{n-1}(A + I)$, for all $n \in \mathbb{Z}^+$.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



7-(MAT/10_HL_Winter_2013_Q4)-Matrices

[Maximum mark: 5]

Consider the matrix $A = \begin{pmatrix} 1 & a(a+1) \\ 1 & b(b+1) \end{pmatrix}$, $a \neq b$. Given that A is singular, find the value of $a+b$.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



8-(MAT/12_HL_Summer_2014_Q3)-Matrices

- (a) Show that the following system of equations has an infinite number of solutions. [2]

$$\begin{aligned}x + y + 2z &= -2 \\ 3x - y + 14z &= 6 \\ x + 2y &= -5\end{aligned}$$

The system of equations represents three planes in space.

- (b) Find the parametric equations of the line of intersection of the three planes. [3]

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....



9-(MAT/12_HL_Summer_2016_Q1)-Matrices

[Maximum mark: 6]

The following system of equations represents three planes in space.

$$\begin{aligned}x + 3y + z &= -1 \\x + 2y - 2z &= 15 \\2x + y - z &= 6\end{aligned}$$

Find the coordinates of the point of intersection of the three planes.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

